Niklas M. Melton

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Education

2017 – 2026 **Ph.D., Computer Science**, 3.6 GPA, Missouri University of Science and Technology
Thesis title: *Incremental Cluster Validity Indices and Their Role in Interpreting Lifelong Learning Systems.*

2012 – 2017 **B.S. Aerospace Engineering**, 3.2 GPA, Missouri University of Science and Technology

Employment History

Graduate Research Assistant, Missouri S&T
Aug 2024 – Present; Aug 2017 – Aug 2021

- Researched machine learning methods for lifelong learning, adaptive AI systems, and LLM-enabled workflows, with publications spanning reinforcement learning, supervised learning, unsupervised learning, and deep learning.
- Executed large-scale experiments on HPC compute clusters, managing training workflows, experiment configuration, and reproducible evaluation pipelines.
- Championed ongoing development of a university-wide RAG system, helping shape retrieval, grounding, and knowledge integration workflows for LLM applications.
- Supported technology transfer efforts with DARPA L2M and the U.S. Army Night Vision Laboratory, translating research into deployable technical capabilities.
- Mentored junior researchers on project design, implementation, and experimental methodology, contributing to multiple publications.

Senior Blockchain Data Scientist, OpenBlock Labs
Apr 2024 – Aug 2024

- Owned the derivatives incentives product, defining program design, analytics, experimentation strategy, and operational delivery for blockchain derivatives clients.
- Led a multidisciplinary team to build and operate production incentive systems for 24/7 cryptocurrency derivatives markets exceeding $200M in daily trading volume.
- Built and maintained production APIs and data pipelines for ingesting, processing, validating, and serving market and rewards data.
- Developed company-wide dashboards to monitor production systems, trading activity, and incentive performance, enabling real-time operational visibility and decision-making.
- Drove MLOps and DevOps practices for model and pipeline deployment, CI/CD testing, and production reliability across analytics systems.
- Managed automated distribution of more than $100M in cryptocurrency incentives; programs contributed to a 5x increase in client trading volume.
- Served as the company's derivatives subject-matter expert in client engagements, translating ambiguous trading requirements into production data systems.

Senior Data Scientist, Janus Health
Mar 2022 – Apr 2024

- Owned *Observe*, a computer vision product for extracting patient and payer data from real-world screenshots of healthcare documents.
- Led a two-person applied machine learning team to develop, deploy, and maintain the end-to-end extraction pipeline in production.
- Increased accurate field extraction coverage from 20% to 98% through iterative model development, pipeline refinement, and validation.
- Applied NLP models (embedding-based clustering and classification) to extracted text to group, categorize, and analyze patient and payer data, enabling structured insights into user behavior and workflow patterns.
- Designed and implemented A/B testing for the *Observe* pipeline to evaluate model and workflow changes in production and guide system improvements.
- Built internal dashboards to track model performance, data quality, and user interactions in production, supporting continuous monitoring and iteration.
- Championed MLOps and DevOps practices including CI/CD testing, model and pipeline management, production deployment, and ongoing performance monitoring.

- Introduced LLM-based pull request review workflows using OpenAI API to accelerate engineering delivery and improve development efficiency.
- Developed production machine learning systems to optimize insurance workflows and predict payout probability.

Senior Data Scientist, Tuscan Services
Aug 2021 – Mar 2022

- Served as the sole data scientist and data engineer, owning the loan scoring and underwriting analytics stack end to end.
- Built production APIs and data pipelines for ingesting customer financial data, managing model workflows, regulations, and predicting loan repayment likelihood.
- Developed dashboards for company-wide use to monitor underwriting models, application flow, and decision outcomes in production.
- Led deployment, CI/CD testing, and model lifecycle management for underwriting and decisioning systems in a regulated environment.
- Enabled processing of 2,000 daily applications and loan approvals totaling more than $50M.
- Maintained model and decisioning compliance with federal and state lending regulations, with attention to validation, auditability, and data governance.
- Replaced rules-based expert systems with machine learning models that improved repayment rates by 30% and increased profit by 10%.

Lead Artificial Intelligence Engineer, Guise AI
May 2017 – Aug 2021 (concurrent with doctoral studies)

- Led AI engineering efforts for computer vision and automated data collection products from prototyping through deployment.
- Managed a four-engineer team developing face and body recognition tools for production use cases.
- Directed MLOps and DevOps workflows including CI/CD testing, model deployment, pipeline maintenance, and production support.
- Designed an intelligent web crawler for automated webpage scraping and content categorization, supporting downstream machine learning pipelines.
- Directed contracted annotation teams to build a human image dataset exceeding 10M labeled samples.
- Worked directly with customers to present technical capabilities, define requirements, and shape project scope.

Skills

Languages	English, B1 German
Coding	Python, C++, R, MATLAB, LaTeX.
ML Libraries	OpenCV, Pytorch, TensorFlow, SciPy, Keras, pandas, Polars, scikit-learn, PyTesseract, LangChain, LangGraph, OpenAI SDK, Anthropic SDK
AI Expertise	Natural Language Processing (NLP), Generative AI, Deep Learning, Reinforcement Learning, Unsupervised Learning, Clustering, Supervised Learning, Large Language Models (LLMs), AI Agents, Decision Trees, XGBoost, Linear Regression, Causal Inference, Explainable AI, Transfer Learning, Computer Vision, Diffusion Models, Fine-Tuning, Transformers, ChatGPT, Claude, Retreival Augmented Generation (RAG)
Databases	MySQL, PostgreSQL, HSQL, SQLite, RDS.
Dev/ML Ops	Git, Docker, Athena, S3, EC2, Poetry, Glue, CloudFormation, SageMaker, EKS, ECS, Spark, Airflow, GitHub Actions, Jenkins.
Misc.	Academic research, training, consulting, LaTeX typesetting, and publishing.

Open Source Projects

2024 **Adaptive Resonance Library (ARTlib)** — Modular Python library for Adaptive Resonance Theory algorithms. GitHub; PyPI; JOSS

2025 **PyAlphaShape** — Python library for alpha shapes in Euclidean and spherical geometries. GitHub; PyPI

2026 **OverlapIndex** — Incremental cluster validity index for quantifying overlap between classes or clusters. GitHub; PyPI

Awards

2025 **Outstanding Paper Award**, "An extensive analysis of match-tracking methods for ARTMAP," IEEE Symposium on Computational Intelligence in Health and Medicine (CIHM).

Best Presentation Award, The need for more nuanced metrics of catastrophic forgetting," in 2025 12th International Conference on Soft Computing & Machine Intelligence (ISCMI).

Research Publications

Journal Articles

1. N. M. Melton, D. Tanksley, and D. C. Wunsch, "Adaptive resonance lib: A python package for adaptive resonance theory (art) models," *Journal of Open Source Software*, vol. 10, no. 114, p. 7764, 2025.
2. N. M. Melton, S. A. Petrenko, and D. C. Wunsch, "Meta-icvi: Ensemble validity metrics for concise labeling of correct, under-or over-partitioning in streaming clustering," *IEEE Access*, vol. 12, pp. 11 114–11 124, 2023.
3. L. E. B. Da Silva, N. M. Melton, and D. C. Wunsch, "Incremental cluster validity indices for online learning of hard partitions: Extensions and comparative study," *IEEE Access*, vol. 8, pp. 22 025–22 047, 2020.
4. R. Imperial, Z. Ahmed, O. M. Toor, C. Erdoğan, A. Khaliq, P. Case, J. Case, K. Kennedy, L. S. Cummings, N. Melton, et al., "Comparative proteogenomic analysis of right-sided colon cancer, left-sided colon cancer and rectal cancer reveals distinct mutational profiles," *Molecular cancer*, vol. 17, no. 1, p. 177, 2018.
5. R. Imperial, Z. Ahmed, O. M. Toor, A. Khaliq, N. Melton, P. Case, J. Case, L. S. Cummings, K. F. Kennedy, S. Hassan, S. Q. Ye, R. M. Mohammad, A. Hussain, T. J. Pluard, J. Subramanian, and A. Masood, "Distinct somatic alterations in right- versus left-sided colorectal cancers.," *Journal of Clinical Oncology*, vol. 36, no. 15_suppl, pp. 3592–3592, 2018.

Conference Proceedings

1. N. M. Melton, L. E. B. da Silva, and D. C. Wunsch, "New metrics for disambiguating feature overlap and catastrophic forgetting in incremental learning contexts," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 40, 2026, pp. 41 319–41 321.
2. N. M. Melton, S. A. Petrenko, J. Liu, J. Schroll, M. Renfrow, I. Sandjaja, and D. C. Wunsch, "A lightweight binary art model for resource-constrained online learning," in *Proceedings of the International Joint Conference on Neural Networks (IJCNN)*, IEEE, 2026.
3. J. Schroll, J. Liu, N. M. Melton, S. A. Petrenko, M. Renfrow, I. Sandjaja, and D. C. Wunsch, "Bfamod: Fpga acceleration of binary fuzzy art for rapid inference on embedded mpsocs," in *Proceedings of the International Joint Conference on Neural Networks (IJCNN)*, IEEE, 2026.
4. N. M. Melton, S. A. Petrenko, L. E. B. Da Silva, and D. C. Wunsch, "The need for more nuanced metrics of catastrophic forgetting," in *2025 12th International Conference on Soft Computing & Machine Intelligence (ISCMI)*, IEEE, 2025, pp. 1–5.
5. N. M. Melton, L. E. B. da Silva, and D. C. Wunsch, "An extensive analysis of match-tracking methods for artmap," in *2025 IEEE Symposium on Computational Intelligence in Health and Medicine (CIHM)*, IEEE, 2025, pp. 1–8.
6. N. Melton and D. C. Wunsch, "Gene expression analysis using adaptive resonance theory," in *Proceedings of the IEEE Computational Intelligence in Bioinformatics and Computational Biology conference (2018, St. Louis)*, 2018.
7. N. Melton and D. C. Wunsch, "Enhancing supervisory training signals with environmental reinforcement learning using adaptive dynamic programming and artificial neural networks," in *2016 IEEE 15th International Conference on Cognitive Informatics & Cognitive Computing (ICCI* CC)*, IEEE, 2016, pp. 331–335.

Preprints

1. N. M. Melton, L. E. B. da Silva, S. Petrenko, D. Wunsch II, et al., *Deep artmap: Generalized hierarchical learning with adaptive resonance theory*, 2025.